

WASP-12b

R-1746

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160225 · created 2026-08-02T19:15:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457443.8049 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.043
Transit depth (Rp/Rs)^2	2.5e-05 +/- 0.0043 [%]
Orbital Inclination (inc)	80.13 +/- 2.89
Airmass coefficient 1 (a1)	1.15 +/- 0.77
Airmass coefficient 2 (a2)	0.072 +/- 0.075
Scatter in the residuals of the lightcurve fit is	65.69 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.0
Transit Duration (day)	0.104 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.043 vs 0.1178 (deviation 1.73 σ)
Duration fit vs published	0.104 d vs 0.125 d (deviation 1.11 σ)
Mid-time O-C	-0.4 min
Depth SNR	0.0
Residual scatter	65.69 %

Light curve

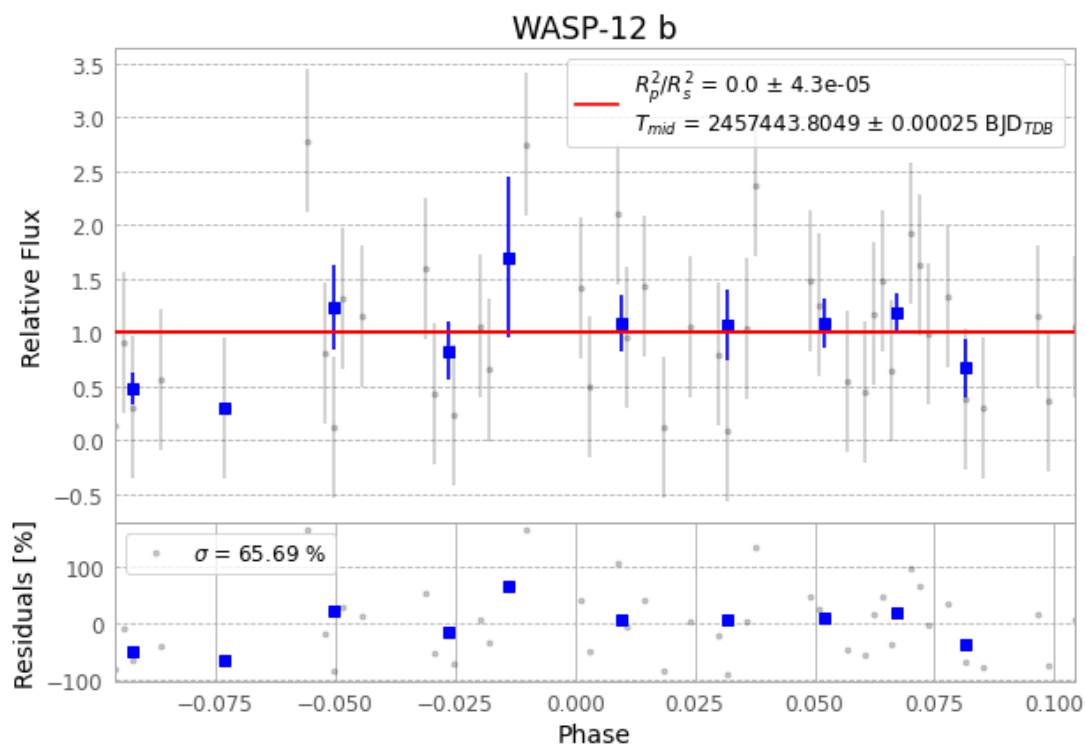


Figure 1 — FinalLightCurve_WASP-12 b_2016-02-24.png (SHA-256 7b52524eca02a035...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [331, 335], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 438e3c9983d7b173...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

108 FITS frames (71 MB), WASP-12160225044212.FITS → WASP-12160225100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160225.log (a0146f3c399c...), FinalLightCurve_WASP-12 b_2016-02-24.png (7b52524eca02...), FinalLightCurve_WASP-12 b_2016-02-24.csv (c9b46c25981f...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 19:15Z.